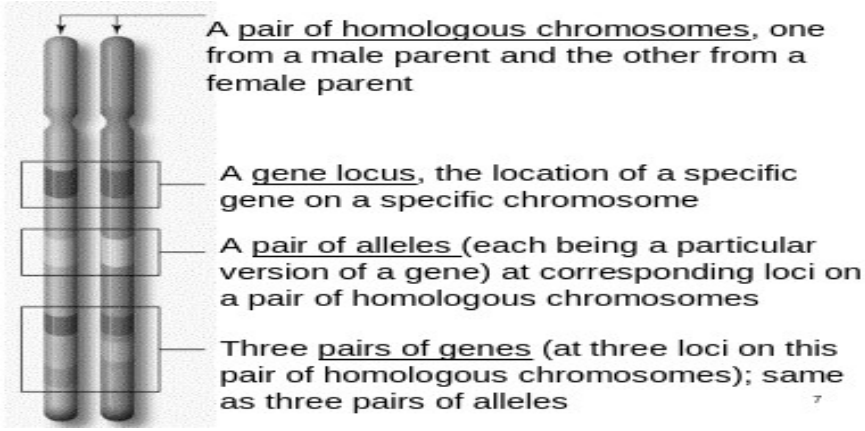
GENETICS

Definitions to be familiar with

- variation, gamete, zygote, fertilisation, chromosome, karyotype, gene, allele, dominant, recessive, homozygous, heterozygous, pure breeding, genotype, phenotype, trait, characteristic, phenotype ratio, Punnett square, pedigree chart, and semi-conservative.

Genetic code

- Cells are the _____ of all living organisms
- Chromosomes are stored in a cells _____
- Chromosomes are made of _____ and contain lots of genes
- Genes are the individual _____ ("words") that are the instructions for _____ and maintaining a living cell
- One gene, for example, will code for _____, another for _____.
- We all have _____ genes for the same feature but we may have different versions of these features.
- Genes for _____ of the same feature are called alleles.
- Everyone has _____ alleles for eye colour – this may be one for blue and one for brown.



- As an example lets imagine a person having one of each from their parents.
- One of these will be dominant and will show through – and the person would have _____ eyes.
- This person would have the ability to pass on either the brown-eyed _____ or the blue-eyed _____ to each of their offspring.
- Each person has _____ chromosomes arranged in _____ homologous pairs

Chromosomes and the Library

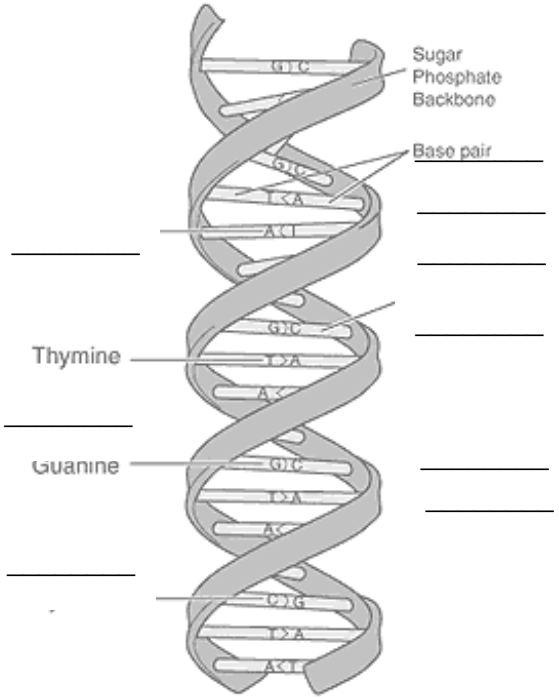
- The chromosomes in a cell's _____ are like a library – they contain a lot of _____ – most of which is useful and essential.
- Each _____ is like a chapter in a book.
- Each set of three bases (called a triplet) is like a _____ in the chapter that refers to something particular.

- A _____ is a change in the sequence of bases and is like a _____ in the book that changes the meaning. Eg **the hot cat** could mutate to become

DNA structure

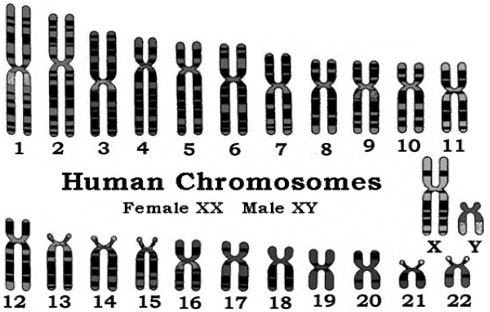
- DNA has a double helix shape (ie a _____ _____)

- DNA is a _____ of sugar, phosphate, and a nucleotide base



Base pairing.
The four bases always pair up with only one other base.
A to
T to
C to
G to

Human karyotype
A diagram which shows all the chromosomes in a cell arranged in their homologous pairs.



Obtained by photographing the chromosomes in the nucleus of a cell.

- Every person has _____ autosomes - non-sex chromosomes.

- Males have an _____ sex chromosome.

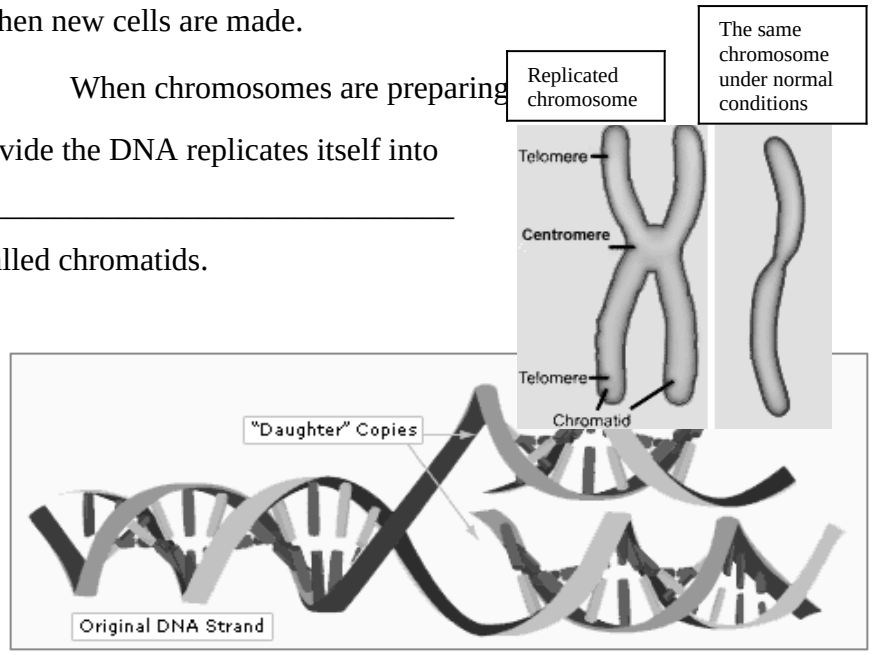
- Females have _____ sex chromosomes

DNA replication

- DNA is unique as a molecule in being able to _____ of itself in a process called DNA replication.

- This happens just _____ when new cells are made.

- When chromosomes are preparing to divide the DNA replicates itself into _____ called chromatids.



Steps in replication

- The double helix _____ apart

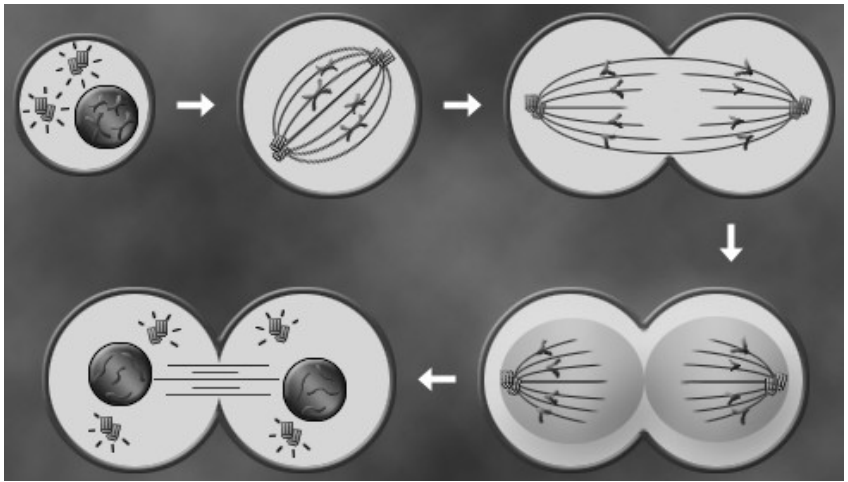
- Enzymes _____ nucleotides according to the base pairing rules.
- The new chromosomes are _____ and are called sister chromatids

Cell division - mitosis

- Purpose - To make _____ for growth, repair and replacement.
- Occurs in _____ (somatic cells)

Sequence of events in mitosis:

1. Chromosomes _____
2. Chromosomes _____ along cell 'equator'.
3. Sister chromatids _____ to opposite sides of cell by spindle fibres
4. Cell _____ forms to separate daughter cells.



Two identical _____ cells produced – in humans, each has _____ chromosomes containing _____ copies of all the genes – the same as the parent cell.

Cell division - meiosis

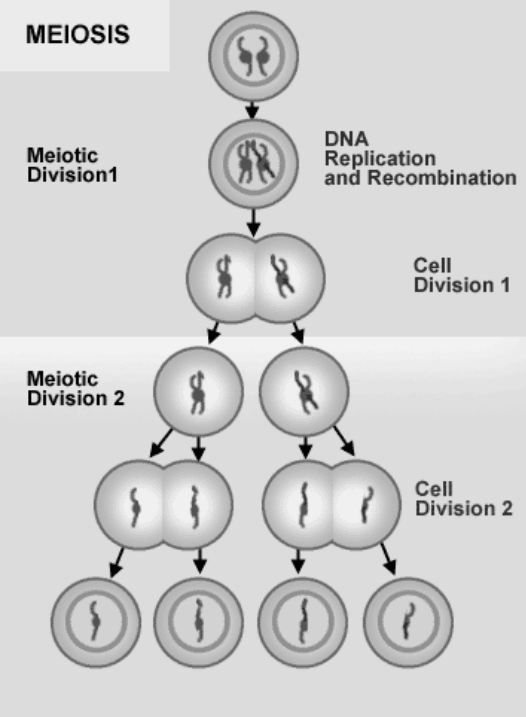
Purpose – to make _____ (sex cells) for reproduction.

Gametes = _____ (eggs) and _____
Sperm are made in the _____ while ova are made in the _____.

Sequence of events in meiosis:

1. Chromosomes _____
2. Homologous chromosomes _____ across cell equator.
3. Homologous chromosomes _____.
4. _____ forms to make separate cells.
5. Chromosomes line up across cell _____.
6. Sister _____ separated
7. New _____ forms.

_____ different _____ cells produced – in humans, each has _____ chromosomes – _____ the number of the parent cell.

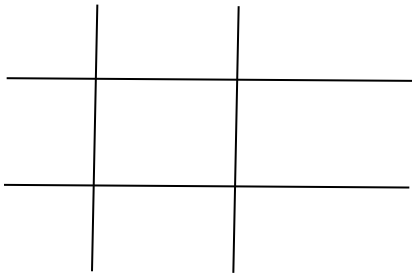


Monohybrid crosses

- A _____ gene is always expressed.
- A _____ gene will be hidden by a dominant gene and will only be expressed when _____.
- _____ letters represent dominant genes; _____ for recessive.
- In _____, tongue rolling (R) is _____ over non-tongue rolling (r).
- Homozygous dominant = _____
- Homozygous recessive = _____
- Heterozygous = _____

E.g. 1 In a family the parents could be:

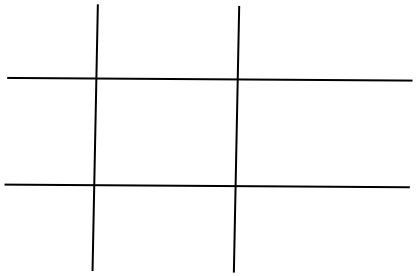
- Father's genotype = _____
- Father's phenotype = _____
- Mother's genotype = _____
- Mother's phenotype = _____
- Father's _____ contain R or r.
- Mother's gametes can only contain _____.
- A Punnett square is used to _____ the expected offspring's genotype and phenotype ratios.



- Therefore _____ of the offspring are expected to be _____; _____ are expected to be _____

- Therefore _____ of the offspring should be able to roll their tongue; _____ cannot.

E.g. 2 If D = dark hair and d = blonde hair, is it possible to get blonde children if the parents are DD and dd?



Therefore _____ offspring will be Dd which means all children will have _____. No children will have blonde hair.